

Principled Alternatives to Naïve Odds Features

Bookmakers' odds carry rich information and should be treated more carefully than "just another feature." One formal approach is to use odds as a **prior** or baseline in a probabilistic model. For example, Egidi *et al.* (2018) build a hierarchical Bayesian Poisson model in which each team's scoring rate is a **convex combination** of parameters learned from historical data and from market odds ¹. In effect, the model treats the odds as a prior estimate of team strengths and shrinks data-driven estimates toward that market-informed baseline. Similarly, Baio & Blangiardo (2010) use hierarchical Bayesian models for soccer goals that pool information across teams to avoid overfitting; these models naturally account for team-level heterogeneity without needing a bivariate correction ².

Another strategy is to incorporate odds as an **offset** in a Poisson framework. For instance, one could convert moneyline odds to implied goals (or win-probability) rates and include that as a log-offset term in a Poisson regression for scores. This forces the model's baseline rate to match the market, and then the remaining features explain deviations. Likewise, odds can serve as an **anchor or separate forecaster** in a stacking ensemble: treat the implied market probability as one "expert" prediction and stack it with other model outputs. In a Super Learner or stacked generalization framework, one trains a meta-model on out-of-fold predictions from the market odds (as one base model) plus other predictors ³. This ensures the market influence is learned and cross-validated rather than simply entered on training data.

A useful alternative is **residual modeling**: predict not the raw outcome, but the residual after subtracting the market's implied probability. In this setup, the target becomes $y - P_{\text{market}}$ or $\text{logit}(y) - \text{logit}(P_{\text{market}})$, and one fits a model to explain any systematic bias left over. This "partialling out" forces the model to focus on what the odds do *not* capture. Koning & Zijm (2022) implement a related idea by regressing actual results on normalized implied probabilities and allowing for match-specific adjustments; their method can include covariates to capture any residual favorite/longshot bias ⁴. In practice, one might calibrate market probabilities (e.g. via Shin's method) and then fit a logistic or Poisson model on the outcomes using those as an offset or by modeling the residual difference.

Market Information as Prior, Offset, Anchor, or Residual

Prior or Convex Combination: In Bayesian models, odds can enter as priors on team strengths or scoring intensities ¹. For instance, set each team's attack/defense parameter to lie between the market-implied value and the historical data estimate. Egidi *et al.* explicitly combine a data-driven Poisson mean with one derived from odds via a convex weight ¹. Similarly, Bayesian network models (Costa *et al.* 2013) can include "subjective" inputs like market odds at higher levels of the hierarchy, updating them with data. These priors anchor the model so that when data are limited, the odds dominate, but when plenty of data accrue, the model can deviate.

Offset in Poisson models: One can use the odds-implied expectation as a fixed offset. For example, convert the moneyline into an implied home-team scoring rate (or win probability) and include its log as an offset in a Poisson GLM for goals ². The model then effectively predicts the *relative* increase/decrease in scoring given other features. This is mathematically akin to modeling $G_i \sim \text{Poisson}(\exp[\log(\lambda^{\text{market}}_i) + x_i^{\text{top}}\beta])$. It formalizes using the market level for each game as a baseline, with the model explaining only the residual.

Calibration Anchor or Separate Forecaster: In ensemble or calibration contexts, treat the market probability as one input to a meta-model. For instance, a logistic meta-learner can take as input (1) the base model's win probability, (2) the Poisson goal-model probability, and (3) the market-implied probability, then learn the optimal blend (possibly with a beta calibration step at the end). Crucially, this stacking must use *out-of-fold* predictions for each component to avoid leakage ³. The SportsStacked example shows selecting top learners and using 5-fold CV to train a neural-network meta-learner that combines them ³. We should do the same with market odds as one “learner.” This design lets the data learn how much to trust the market vs other sources.

Residual Target Modeling: One can explicitly model the residual between reality and market prediction. For win probabilities, define a new target $r = \mathbf{1}\{\text{home win}\} - P_{\text{market}}(\text{home win})$, and fit a regression or classification model on r . This forces zero-mean learning (market already explains most variation), so any nonzero signal is by definition value over market. Similar ideas arise in logistic regression with an “offset” logit of the market. The 2022 Koning & Zijm study essentially shows this: they regress outcomes on normalized odds plus other features, capturing any remaining bias ⁴. Their findings (e.g. Shin-adjusted odds are unbiased in EPL but not in La Liga) imply one should test whether such residuals exist in NRL.

Methods for Small-to-Medium Data with Many Features

Given limited data, avoid overly complex models. Bayesian or regularized approaches help. Hierarchical Poisson models (Baio & Blangiardo 2010) pool team effects to reduce variance ². Likewise, imposing structured shrinkage (e.g. ridge or hierarchical priors) can prevent any one feature (like odds) from dominating, unless truly predictive. In machine learning, use strong regularization or monotonicity constraints on the odds feature to prevent “swamping” (e.g. constrain its coefficient in a GLM or tree ensemble). A small data regime also suggests favoring simple models: e.g. combine an interpretable Elo-like rating or attack-defense model (potentially dynamic through the season) with a lightweight residual learner.

Time-order requires *forward* validation. Use rolling-origin or forward-chaining cross-validation rather than random splits. For example, train on seasons up to 2022 and test on 2023–2025, or use k-folds that respect chronology (each fold's training data precedes its test data). This avoids lookahead: odds and features in the test set must only use information up to that point. Avoid “leaking” future features by timestamping. For any feature engineered from form or context, ensure it uses only pre-game data.

Feature selection or dimensionality reduction can also help. With many engineered stats, consider L1/L2-penalized logistic/Poisson regression or tree methods with early stopping. Bayesian model averaging or

stacking (superlearner) can balance linear strength models (e.g. team-strength models) with complex ML models. In practice, blending a simple parametric model (which is more data-efficient) with a GBM or network “residual” model often works best on moderate data sizes.

Score Models vs. Win-Probability Models in Team Sports

In low-scoring sports (soccer, rugby), modelling goals or tries by Poisson/Dixon-Coles is common. Rugby league/union have discrete scoring but similar structure, so Poisson or bivariate-Poisson processes apply ⁵. Tonkes (2016) derives NRL scores using Poisson processes with time/score-dependent rates calibrated to both historical scores and bookmaker data ⁵. In contrast, NFL and basketball (higher scoring) often use continuous approximations or logistic models on score differential. However, one can still use an independent-Poisson approach for scores or a Gaussian for margins.

For win probabilities, Elo or Bradley–Terry models are standard across sports (soccer, rugby, NFL). These models provide a baseline “league-average” win probability that can be updated with factors. They can feed into a residual ML model (dynamic hierarchical strength → residual learner). Any sport-specific nuance (e.g. field-goal frequencies in NFL vs tries in rugby) should inform the score model but does not fundamentally change the market-integration approach.

Comparing Modeling Approaches

To compare strategies, set up controlled experiments on hold-out data (2023–2025). For each approach, train on 2010–2022 and evaluate non-draw log-loss, calibration, and economic metrics. Key variants to test:

- **Odds as raw features:** include moneyline/log-odds or implied probability directly in the feature vector of a Poisson or logistic model (baseline).
- **Odds as priors/offsets:** fit a Poisson model for scores with an offset term equal to the log of the market-implied goal rate, letting other features have zero mean in that offset space. Alternatively, use odds-implied scoring rates as a Bayesian prior.
- **Residual modeling:** train a model on the residual target $y - P_{\text{market}}$ (or $\log(y) - \log(P_{\text{market}})$), effectively using market as prediction of $y=1$. This requires careful handling of probabilities.
- **Out-of-fold stacking:** create an ensemble where one base “model” is a predictor that outputs the market probability, plus others (team-strength model, LightGBM model, etc.). Use proper k-fold CV for base models, then train a meta-learner on these OOF predictions ³. Compare this to naive in-sample stacking (to confirm the benefit of OOF).
- **Constrained/regularized models:** fit models where the odds feature has a capped coefficient. For example, in a logistic regression one could force the odds-feature weight ≤ 1 . Or in tree-based models use monotonic constraints so that higher odds cannot reduce win probability beyond a limit. Compare performance to unconstrained versions.

For each variant, measure: - *Predictive metrics:* log-loss, Brier score, calibration (reliability curves), possibly RPS for multi-outcome (win/draw/loss).

- *Market-relative metrics:* closing-line value (average improvement in implied odds) and betting return. For

instance, simulate placing bets when the model's probability diverges sufficiently from the market (as in Page 2016 ⁶). Also compare log-loss of the model vs the implied odds baseline: Lopez (2017) shows market log-loss vs model log-loss (-0.6845 vs -0.6836) to gauge lift ⁷. A Diebold–Mariano test on forecast errors vs market probabilities could assess significance.

Evaluation Beyond Predicting Winners

Focus on *value* rather than raw accuracy. The evaluation should simulate betting or measure calibration relative to the market. One measure is **closing-line value (CLV)**: compute for each game the difference between the odds you'd get at prediction time and the closing odds, and see if the model consistently "beats the market." Another is the **expected return on investment**: place hypothetical bets according to the model's edges (e.g. flat or Kelly betting) and track ROI ⁶. As Page (2016) illustrates, even polling models can be tested by "betting" against Betfair prices and measuring ROI ⁶. Similarly, compare the model's log-loss to the market's (as Lopez did), and examine whether gains are robust to vig (house edge) ⁷.

Importantly, use *time-respecting splits* for evaluation: do not shuffle games across seasons. Reserve entire seasons (2023–25) as out-of-sample. Also consider "paired" statistics: for each game, compare the model's predicted win probability vs the market's implied probability in terms of calibration and actual outcome. Computing the difference in log-loss for each game and aggregating can highlight if the model systematically improves on the market.

Leakage and Timing Risks

Be very careful with timestamps. If odds are taken "near kickoff," they may incorporate very late information (injuries, lineup changes, weather). Ensure no feature uses information from after the odds timestamp. For example, if you use a lineup predictor, cut off inputs to only those known at the same time as the odds. Otherwise you risk leaking outcome-related info. Similarly, odds themselves often move up to the last second; if you use the *opening* odds but also use features updated later, you'll give your model unfair advantage. One approach is to fix a single decision time (say 1 hour pre-game) and use features and odds available at that cutoff.

Conversely, using older features (e.g. team form from weeks ago) when odds reflect very recent form might underestimate the market. The safest protocol: align every feature with the official timestamp of the odds you use. Any lag (e.g. using 3-day-old data when odds are current) can bias the residuals. Also beware "lookahead" in engineered features: e.g., rolling averages should be computed only with past games. In practice, logging the data pipeline and verifying no feature accidentally uses future info (data snooping) is essential.

Ranked Strategies (by Lift / Cost / Risk)

1. **SuperLearner Stacking with Market as Base**
2. *Lift*: High. Combines all information, likely best accuracy. Out-of-fold ensures no overfitting.
3. *Cost*: Moderate. Requires implementing full CV stacking pipeline and careful data splits.

4. *Risk*: Low. Proven ensemble method ³ ; transparency manageable if meta-model is simple.

5. Hierarchical Poisson with Market Prior (Convex Mix)

6. *Lift*: High. Combines strengths of data and market; Egidi *et al.* report strong predictive fit ¹ .

7. *Cost*: High. Needs Bayesian modeling (potentially MCMC).

8. *Risk*: Medium. Must set mixing weight or prior strength; risk of mis-specification if market not fully efficient.

9. Poisson with Odds Offset

10. *Lift*: Medium. Uses market as baseline for scores; allows features to explain deviations.

11. *Cost*: Low. Simple GLM change.

12. *Risk*: Low. Statistically principled; risk if implied rates are inaccurate.

13. Residual Classification on Market Probability

14. *Lift*: Medium. Directly learns any edge beyond market.

15. *Cost*: Low. Just transform target.

16. *Risk*: Low. If market is nearly optimal, residual may be small/noisy.

17. Regularized/Constrained Models (Limit Odds Influence)

18. *Lift*: Medium. Helps mitigate over-reliance on odds.

19. *Cost*: Low. Use L1/L2 penalty or monotonic constraint.

20. *Risk*: Low. Might slightly reduce performance if market truly holds most information.

21. Benchmark Odds Forecaster + Meta-Calibration

22. *Lift*: Low-to-Medium. Easier to implement as you calibrate the market probability (e.g. isotonic regression, beta calibration) to remove bias.

23. *Cost*: Low. Just additional calibration step.

24. *Risk*: Low. Won't hurt base predictions much, but may yield limited gain.

25. Dynamic Team-Strength Model + Residual GBM

26. *Lift*: Medium. Fit a dynamic Elo or rating system for base probabilities, then a residual ML model on features.

27. *Cost*: High. Need to develop/validate the rating model.

28. *Risk*: Medium. Gains if data limited (reduces variance), but complex.

29. Market-Implied Spread with Residual Regression

30. *Lift*: Low. Compute implied expected margin from odds, model residual actual margin.

31. *Cost*: Low. Just a linear model on scores.

32. *Risk*: Low. Simple but may underperform direct score models.

33. Segmentation by Market Movement

34. *Lift*: Low. Partition games by how much odds move; train separate models.

35. *Cost*: Medium. Requires handling class imbalance.

36. *Risk*: Medium. Useful if late news is significant, but often noisy.

37. Ignore Odds, Focus on Data (Baseline)

38. *Lift*: Negative (no odds signal used).

39. *Cost*: Medium. Already done.

40. *Risk*: High. Market ignored may lead to poor calibration.

Architecture Recommendations

- **Hybrid Pipeline**: Use a Bayesian or Elo-based model to capture long-term team strengths (Tier A), combined with a LightGBM-like model for short-term/context features (Tier B). Then stack these with market odds (Tier C) in a CV-based meta-model ³.
- **Market Input Modules**: Compute market features in multiple ways: e.g. implied win probability (normalized odds or Shin-adjusted ⁴), implied goals (for Poisson offset), or betting spreads. Use all relevant forms as inputs.
- **Out-of-Fold Blending**: Implement stacking such that for each game, the meta-model only sees out-of-fold predictions of Tier A, B, and market. A simple meta-learner (e.g. logistic regression or shallow GBM) can then weight them.
- **Regularization on Market Weight**: In the meta-model or standalone models, constrain the weight on market-based features (for example, by L2 penalty). This avoids letting odds overshadow everything else.
- **Bayesian Updating Layer**: If feasible, add a Bayesian layer where market odds set a prior, updated with features. For instance, use a Beta-Binomial or conjugate Poisson-Gamma formulation where the odds-derived probability is the prior mean and Tier B features adjust it.

Experimental Designs

1. **Holdout Comparison**: Train each variant (listed above) on seasons ≤ 2022 and evaluate on 2023–2025. Compare non-draw log-loss and Brier scores for home-win vs away-win. Report differences against the pure market baseline (simply using implied probs).
2. **Betting Simulation**: For each model, simulate bets by staking on games where the model's win probability exceeds market by some margin (e.g. $\geq 5\%$). Track cumulative profit and ROI, accounting for vig. Use a fixed-kelly strategy or flat stake to compare strategies ⁶.
3. **Ablation Tests**: Remove odds-related features to see how much performance drops. Likewise, try using only market odds (as a null model) to set a floor.

4. **Cross-Validation for Stacking:** Verify that OOF stacking outperforms in-sample stacking (train meta-model on out-of-fold predictions vs on training predictions). This prevents target leakage in the stack.
5. **Shin vs. Normalization:** Experiment with two ways to turn odds into probabilities – simple inverse normalization vs Shin’s adjustment ⁴ – as inputs or priors. Choose whichever yields better calibration.

Modeling Market Information

Given the results to date, I recommend modeling market data mainly as priors or offsets rather than unrestricted features. Specifically:

- **Prior on Scoring Intensity:** Use the bookmaker’s expected goals (from odds) as a Bayesian prior or offset for each team’s Poisson rate. Then let features only adjust that prior. Egidi *et al.*’s convex-combination approach ¹ is a template.
- **Offset in Poisson GLM:** Implement a Poisson regression for home/away scores with `offset = log(\lambda_{market})`. Features then model `\lambda_{actual} = \lambda_{market} * exp(x \beta)`. This ensures the baseline exactly matches the market expectation.
- **Residual Win Probability:** In the win-probability model, include logit(implied probability) as an offset or as the sole predictor plus features. In essence, predict the residual log-odds beyond the market.
- **Separate Forecaster in Stack:** In the meta-stack, include one expert that always outputs the market probability (or a simple calibrated version). Let the meta-learner reduce its weight if the data suggest so.
- **Calibration Anchor:** As a final step, consider recalibrating combined predictions against the market baseline (e.g. blend the final probability slightly toward the market closing line). This avoids extreme predictions unsupported by the market.

Preserving Incremental Signal

To prevent odds from dominating: cap their influence with regularization or constraints, and ensure other features get a chance to adjust. For example, train the model on the residual (market removed) so that only unexplained variation remains. Alternatively, if using a GBM or neural net, downweight or narrow the range of the odds feature (e.g. by using rank or a smaller scaling) so that splits must earnestly justify using it. In a meta-learner, impose a prior or penalty making the meta-weight on the market model at most 0.8, forcing some credit to other models.

Finally, continually check feature importance: if odds still register as overwhelmingly the top predictor, tighten the constraint (increase penalty or reduce learning rate for that branch). The goal is “efficient use” of market signal, not suppression, so some weight is appropriate, but it should not overshadow robust historical and situational data.

Sources: Recent sports analytics and forecasting literature ¹ ⁴ ⁷ ³, as well as market-efficiency studies (e.g. Koning & Zijm 2022) support these strategies. These references describe hierarchical Poisson models with market priors, stacker designs with OOF predictions, and calibration methods for odds, all of which are directly relevant to improving the NRL pipeline.

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